

Basic Disk Imaging using FTK Imager

1. Introduction

Disk imaging is an important process in digital forensics used to create an exact copy of a storage device or selected data. This ensures that digital evidence is preserved without altering the original source. In this task, FTK Imager was used on Windows OS to create a forensic image and generate hash values for verification.

2. Objective

- Understand the forensic acquisition process.
 - Create a forensic image using FTK Imager.
 - Generate and verify MD5 and SHA1 hash values.
 - Preserve digital evidence integrity.
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3. Tools & Platform Used

Tool: FTK Imager
Operating System: Windows OS
Target Source: USB drive / Sample Folder

4. About FTK Imager

FTK Imager is a forensic acquisition tool developed by AccessData. It is commonly used to capture forensic images of drives, preview files and folders, recover deleted files, generate hash values, and preserve evidence integrity.

5. Target Selection

The selected target for imaging was a USB storage device or sample folder containing test files.

6. Imaging Process

Step 1: Launch FTK Imager on Windows OS.
Step 2: Select source using File → Create Disk Image.
Step 3: Choose the USB drive or folder as evidence.
Step 4: Select image format (E01 or DD).
Step 5: Add case information.
Step 6: Select destination folder.
Step 7: Enable MD5 and SHA1 verification.
Step 8: Start the imaging process.

7. Image File Created

Example image files generated:

- evidence.E01
 - evidence.dd
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8. Importance of Hash Verification

Hash verification ensures evidence integrity, authenticity, and confirms that the forensic image has not been modified during acquisition.

9. Conclusion

The task successfully demonstrated the forensic imaging process using FTK Imager. A forensic image was created and verified using MD5 and SHA1 hash values, ensuring digital evidence integrity during forensic investigations.

Hash Values Generated

Hash Type	Example Value
MD5	d41d8cd98f00b204e9800998ecf8427e
SHA1	da39a3ee5e6b4b0d3255bfef95601890afd80709

10. Additional Comments

- FTK Imager is widely used in digital forensic investigations.
 - Imaging should be performed carefully to avoid altering evidence.
 - E01 format supports compression and metadata storage.
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